

Team Control Number
510105201001220151

Problem Chosen
Wildlife Protection
2026
IM²C
Summary Sheet

Planning Wildlife Protection under Geographic and Resource Constraints

Large reserves need inspections that lead to timely action. We connect conservation priorities, ground response and limited personnel and drones in Etosha, then examine seasonal work and transfer to other parks.

Priorities. We address illegal entry and poaching, harmful fire and habitat damage through a multi-species framework. Six animal groups with usable 2015 survey records and burnable-habitat areas define relative responsibility. The Effective Inspection Completion Rate measures prescribed checks supported by timely ground response.

Allocation. A two-stage linear program first maximizes completion, then minimizes labor across all equally optimal service patterns. With **21240** personnel-hours and **1200** flight-hours available monthly, its **57.26%** optimum uses **5286.11** personnel-hours and **144.34** flight-hours. Response access limits service. The best of three additional-post candidates raises completion to **64.27%**.

Seasonal staffing. Separate fire checks and maintenance of 17 historical boreholes require **57** personnel equivalents in the normal peak month and **62** under the specified drought-maintenance rule, at 120 effective hours per person-month.

Sensitivity. All three allocation rules tie under ample resources. At a 3600-hour labor limit, uniform, demand-based and optimized completion are **16.47%**, **19.87%** and **29.88%**. Reduced drone efficiency restores ground screening; grid size changes the estimated response ceiling.

Evaluation. Independent checks confirm model feasibility and agreement between allocation and inverse-workforce calculations. Historical records and assumed protocols limit operational interpretation; service completion does not establish wildlife survival or sufficient ecological protection.

Transfer. Chitwan and Yellowstone trials rebuild local geography. A historical Yellowstone wolf-use scenario changes priorities and deployment. Chitwan's published geometry and official area remain inconsistent, so its results are conditional. Validate access and task effectiveness before adopting staffing recommendations.

Keywords: wildlife protection; resource allocation; ground response; seasonal workforce; Etosha National Park.

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Reading guide. Sections 4–9 answer Questions 1–6. The letter uses nontechnical language. References and the AI-use report are included within this document’s 25-page ceiling.

1. Introduction

Large wildlife reserves are difficult to protect because important areas are spread over a very large space, while staff, equipment and travel time are limited. Observations are useful only when qualified teams can reach the location and respond in time. We use Etosha National Park as our main case, with a reported area of about 22,935 km² and a supplied workforce reference of 295 employees [1]. We ask how different areas should be prioritized, how ground teams and drones should be allocated, and whether the same service level can be maintained when seasonal work increases.

Our model divides the park into planning units and combines historical animal data, burnable habitat and mapped access with inspection tasks. An inspection receives effective-service credit when its required screening is completed and a ground team can arrive within two hours. The Effective Inspection Completion Rate measures the weighted inspection work supported in this way. It does not directly represent wildlife survival or continuous coverage of the whole park. Six surveyed animal groups enter the quantitative model; missing species and current animal distributions still require field data.

We build a two-stage linear programming model. The first stage finds the highest completion rate under the stated constraints, and the second minimizes personnel-hours while preserving that rate. For seasonal work, we add fire screening and maintenance of 17 historical boreholes. Personnel requirements are estimated from workload using 120 effective working hours per person per month. We vary resources, inspection rules, grid size and weights to examine how the assumptions affect the result. For Chitwan and Yellowstone, we rebuild local geography and priorities instead of directly copying Etosha’s staffing results.

Under the baseline setting, completion reaches 57.26%, using 5286.11 personnel-hours and 144.34 flight-hours. Resources remain unused, so the main constraint is response access rather than the total amount of staff or drones. The busiest normal seasonal period requires 57 personnel equivalents; the drought-maintenance scenario requires 62. These task-based estimates are not the park’s exact staffing needs. Figure 1 gives an overview of the data, models, checks and recommendations.

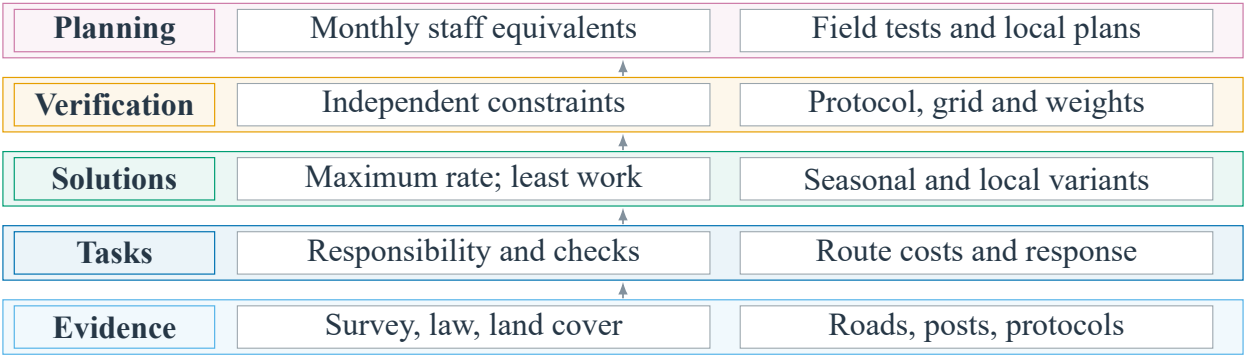


Figure 1: Overview of the data, models, checks and recommendations.

2. Basic Assumptions and Justifications

The following assumptions define a planning scenario rather than verified park operating rules.

Assumption 1: Work is pooled monthly, using 120 effective hours per person, 60% personnel availability and 30 drones.

Justification: Declared limits permit comparable deployment tests; monthly feasibility does not prove a daily roster.

Assumption 2: Inspection demand and within-region responsibility are allocated by area, with equal fire exposure on burnable habitat.

Justification: Fine animal locations and multiyear harmful-fire exposure are unavailable; explicit proxies avoid inventing observations.

Assumption 3: Ground and drone teams complete the same checklist only where suitable; credited service needs ground arrival within two hours and shared teams handle infrequent alerts.

Justification: Observations require follow-up. Task equivalence, service standards and concurrency still need field calibration.

3. Data Description and Symbol Definitions

The problem gives an approximate area of 22,935 km²; our projected boundary covers 22897.10 km². We use 2015 animal surveys, 2021 land cover and a fixed 2026 road snapshot [1, 2, 4, 5], organized into 17 reporting regions and 599 inspection units. A representative location supports route calculations, not continuous area coverage. Question 2 excludes water and rainfall; Question 3 separately adds fire screening and maintenance of 17 historical boreholes without forecasting populations or water-delivery volumes.

Table 1 distinguishes reporting regions, local inspection units and months. Normalized weights represent responsibility, while completion records checklist work supported by ground response. Personnel-hours and flight-hours remain separate: drone operation also consumes personnel work, and response readiness is counted once for the park.

Table 1: Main symbols and units used in the protection-planning model.

Symbol	Definition
i, j, t	Reporting region, inspection unit and month.
A_j, C_j	Unit area and prescribed monthly checks.
w_j, s_j	Relative inspection importance and completion fraction.
P, η	Effective inspection completion rate and target, expressed in percent.
r_j	1 when ground staff can arrive within the time limit; 0 otherwise.
x_j, h_j	Ground screening personnel-hours and drone flight-hours.
H, U, H_0	Personnel limit, flight limit and reserved response work.

4. Protection Priorities and a Measurable Standard

We identify shared threats before assigning regional responsibility. The management goal covers wildlife across habitats; the animal component quantifies the six surveyed groups rather than a complete species inventory.

4.1 Priorities and evidence

Table 2 ranks concerns using threat evidence, potential loss, affected extent and opportunities for intervention. Illegal entry and poaching lead to animal-monitoring and response tasks; harmful fire and habitat disturbance lead to habitat-monitoring tasks. The ranking is a management judgment, rather than measured incident probabilities or an exhaustive risk inventory.

Table 2: Conservation priorities used to define the Etosha inspection tasks.

Priority	Challenge and evidence
1	Poaching and illegal entry: unauthorized access creates enforcement needs across animal habitats; the supplied rhinoceros-poaching context is one documented example.
2	Harmful wildfire and habitat damage: potentially widespread effects on animals and habitats; distinguish prescribed fire.

The animal component uses blue wildebeest, plains zebra, giraffe, African savanna elephant, gemsbok and ostrich. We retain source quality flags and exclude unresolved count tables. Survey blanks contribute no recorded abundance, rather than confirmed zero populations. These historical records support a deployment demonstration, not current abundance for all wildlife. Rhinoceros poaching remains a qualitative concern;

its conflicting counts are not required by this multi-species model. Human–wildlife conflict, disease and species-specific interventions remain outside the quantified task list.

Historical abundance is converted to within-species regional shares, then category preferences determine responsibility. Burnable-habitat area forms a separate component. Equal baseline fire exposure avoids extrapolating a single sample into multiyear harmful-fire risk.

4.2 What effective inspection completion measures

A target receives effective service when prescribed screening is completed and a qualified ground team can arrive within the selected response limit. Let $d_j \in [0, 1]$ describe screening completion and $r_j \in \{0, 1\}$ describe response eligibility. For normalized demand weights w_j , define

$$\begin{aligned} g_j^{\text{eff}} &= d_j r_j, \\ P &= 100 \sum_j w_j g_j^{\text{eff}}, \\ L &= \sum_j w_j (1 - g_j^{\text{eff}}). \end{aligned} \tag{4.1}$$

The allocation uses $s_j \leq r_j$ directly. With fixed weights, larger P means less unmet weighted demand; $P = 57.26$ denotes 57.26%, independently of area coverage or animals protected.

Actual protection needs independent evidence: illegal-loss records with comparable monitoring effort, population surveys, habitat condition and the impact of harmful fires. More detected incidents may reflect better detection. Prescribed burning is not automatically protection failure. A park average cannot certify missing species or replace their dedicated safeguards.

5. Resource Allocation with Ground Teams and Drones

The priorities above determine which inspections receive weight. This section converts their completion requirements into travel, personnel and flight work, then chooses a feasible allocation.

5.1 Planning units and resource assumptions

The 15 survey regions, main salt pan and other survey-external areas form 17 reporting regions $\mathcal{I}$. A 10 km grid intersected with each region produces 599 connected inspection units $\mathcal{J}$. Each has an interior representative location and support area A_j . The location is a route-planning proxy, not proof of continuous coverage of that area.

Table 3 states the operating assumptions. Speeds convert mapped routes into costs and response times; checklist, endurance and openness rules determine where drones can be used. Six sourced locations are assumed response posts. Monthly availability does not mean a fixed fraction of employees must be on duty simultaneously.

Table 3: Baseline operating assumptions for the Etosha allocation model.

Item	Setting
Travel speed	Road 30, walking 4, drone flight 40 km/h.
Task time / team	Observation 10, preparation 5, dispatch 10 minutes; two people per team.
Service / response	8 checks per 100 km ² /month; ground arrival within 2 hours.
Technology eligibility	At most 0.6 flight-hours/check; regional openness at least 60%.
Access / service floor	Entry-index decay scale 2 hours; reachable-area service floor 15%.
Personnel	295 reference employees; 60% availability; 120 hours/person-month.
Drones	30 aircraft; 2 flight-hours/day for 20 flyable days/month.
Readiness	Six candidate posts, two people each, 8 hours/day for 30 days.

The two resource limits and shared response reservation are

$$\begin{aligned} H &= 295(0.60)(120) = 21240 \text{ personnel-hours/month}, \\ U &= 30(2)(20) = 1200 \text{ flight-hours/month}, \\ H_0 &= 6(2)(8)(30) = 2880 \text{ personnel-hours/month}. \end{aligned} \quad (5.1)$$

The available inspection labor before task allocation is $H - H_0 = 18360$ hours. Response readiness is charged once for the park. Staffing fractions, flight allowances and service standards are explicit planning assumptions; they are not verified operating rules at Etosha.

5.2 Animal responsibility, habitat and access

For species m , let N_{im} be its surveyed abundance in region i , $q_{\ell(m)}$ its category weight, and $\mathcal{I}_S$ the 15 surveyed regions. Let A_i^F denote burnable-habitat area. Define

$$\begin{aligned} V_i^A &= \frac{\sum_m q_{\ell(m)} (N_{im} / \sum_{v \in \mathcal{I}_S} N_{vm})}{\sum_m q_{\ell(m)}}, \\ V_i^F &= \frac{A_i^F}{\sum_v A_v^F}. \end{aligned} \quad (5.2)$$

Hunttable, protected and specially protected categories use the team comparison matrix

$$\begin{aligned} \mathbf{A} &= \begin{pmatrix} 1 & 1/2 & 1/3 \\ 2 & 1 & 1/2 \\ 3 & 2 & 1 \end{pmatrix}, \\ \mathbf{q} &= (0.163424, 0.296961, 0.539615)^\top. \end{aligned} \quad (5.3)$$

The normalized principal eigenvector gives $\mathbf{q}$; $CR = 0.007933 < 0.1$ checks consistency, not ecological truth. The law supplies categories, not these numerical comparisons [3].

For unit j , the shortest entry time from one of 27 road–boundary intersections is $T_j^E = D_j^E / v_g + d_j^\perp / v_w$. Its access index is $E_j = \exp(-T_j^E / 2)$ with time in hours. These are potential access sources, not proven poacher entrances. Figure 2 compares demand with area service. Within a region, $\mu_j = A_j / A_i$ apportions responsibility by area: $R_j^A = V_i^A \mu_j E_j$ and $R_j^F = V_i^F \mu_j$.

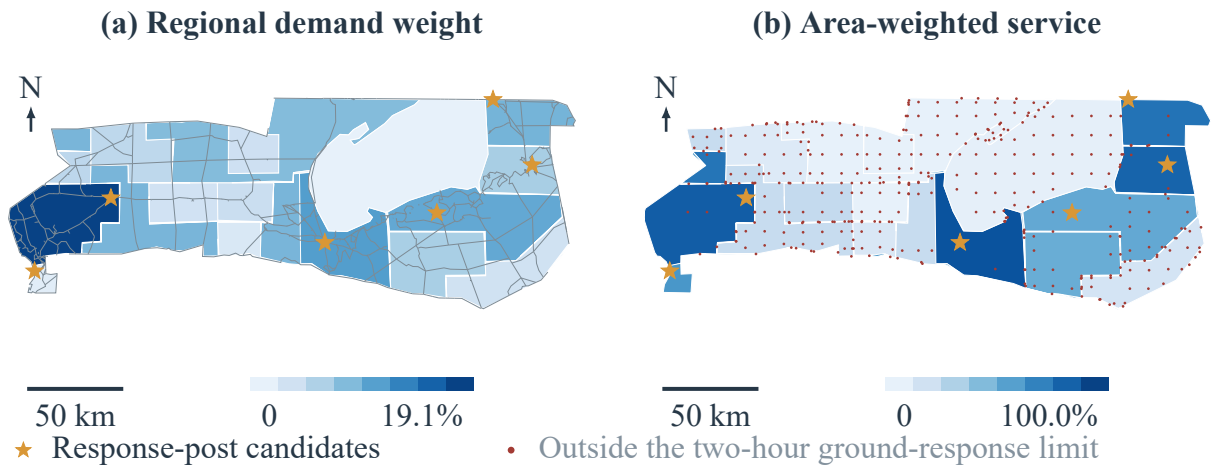


Figure 2: Regional responsibility and response-supported area service in Etosha.

In Figure 2, the demand panel uses historical animal responsibility, burnable habitat and potential entry access; the service panel shows area-weighted completion from the same 599 units. Stars mark the assumed response posts, and red points fail the two-hour response rule. Priority-weighted completion and area service have different denominators. Outside-survey animal responsibility remains unknown; zero measured coefficients and zero salt-pan fuel do not imply zero ecological value.

5.3 Weights, routes and response eligibility

We estimate the two component weights over the common 15-region scope. First aggregate $X_{ik} = \sum_{j \in \mathcal{J}_i} R_j^k$, then calculate

$$\begin{aligned} p_{ik} &= \frac{X_{ik}}{\sum_v X_{vk}}, \\ e_k &= -\frac{\sum_i p_{ik} \ln p_{ik}}{\ln 15}, \\ \alpha_k &= \frac{1 - e_k}{(1 - e_A) + (1 - e_F)}. \end{aligned} \quad (5.4)$$

With $0 \ln 0 = 0$, $e_A = 0.881596$ and $e_F = 0.934989$ give $\alpha_A = 0.645551$ and $\alpha_F = 0.354449$. Entropy weights express spatial differences in the chosen data, not observed loss shares. Each component is separately normalized over its observed scope, then $w_j = \alpha_A w_j^A + \alpha_F w_j^F$ and $W_i = \sum_{j \in \mathcal{J}_i} w_j$. Thus $\sum_j w_j = 1$.

The directed road graph retains original connections and one-way restrictions. Outbound and return distances are calculated separately. A post qualifies only if its round trip is finite and

$$\begin{aligned} t_d + d_{bj}^+/v_g + d_j^\perp/v_w &\leq T_{\max}, \\ r_j &= \mathbf{1}\{\text{at least one qualifying post}\}. \end{aligned} \quad (5.5)$$

We choose the qualifying post with the shortest road round trip $D_j = d_{bj}^+ + d_{jb}^-$. Response eligibility concerns the outbound journey; task cost also includes the return. Authorized use of candidate roads and straight-line off-road travel are assumptions requiring field checks.

$$\begin{aligned} a_j &= n_g(D_j/v_g + 2d_j^\perp/v_w + t_o + t_p), \\ b_j &= 2d_j^\perp/v_u + t_o, \\ o_j &= n_u(D_j/v_g + t_p + b_j), \\ \gamma_j &= o_j/b_j. \end{aligned} \quad (5.6)$$

Ground cost a_j and drone-support cost o_j are personnel-hours per check; b_j is flight-hours per check. Drone use h_j therefore consumes $\gamma_j h_j$ personnel-hours. Only response-eligible units meeting endurance and openness conditions enter $\mathcal{J}_U$. Figure 3 compares an example check at ENP1_0000: ground screening takes 4.981 personnel-hours, while the drone-assisted task takes 4.271 personnel-hours and 0.206 flight-hours. The segments separate travel, walking, observation or flight, and preparation. The saving replaces off-road walking; both tasks retain transport and human involvement. These are modeled costs, so checklist equivalence still needs field testing.

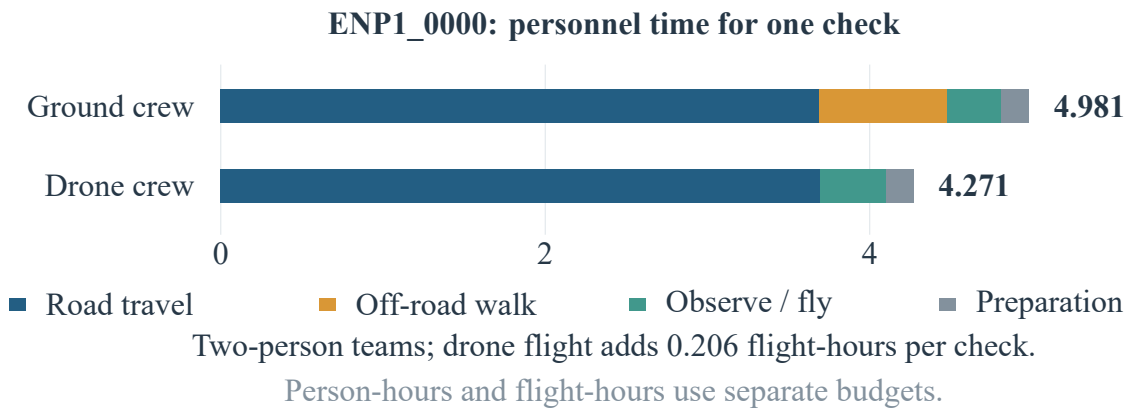


Figure 3: Modeled personnel-hours for ground and drone-assisted inspection at ENP1_0000.

5.4 Service capacity and the allocation model

Decision variables are ground screening hours x_j , drone flight-hours h_j , and effective completion s_j . The prescribed monthly checks are based on area:

$$\begin{aligned} C_j &= 8A_j/100, \\ \sum_j C_j &= 8 \sum_j A_j/100. \end{aligned} \quad (5.7)$$

Areas are in km^2 , giving 1831.77 prescribed checks per month. Refining the grid redistributes area without creating additional total demand. A completed check is a task at a representative location. Fractional checks describe planning intensity and require integer dispatches for implementation.

Define capacity coefficients $g_j = 1/a_j$ for eligible ground tasks, otherwise zero; $u_j = 1/b_j$ for eligible drone tasks, otherwise zero. Then $C_j s_j \leq g_j x_j + u_j h_j$. This avoids dividing by undefined costs and permits different checks to be shared between technologies without double-counting the same inspection.

For each region, $Q_i = \sum_{j \in \mathcal{J}_i} \mu_j r_j$ is reachable area share and $\bar{s}_i = \sum_{j \in \mathcal{J}_i} \mu_j s_j$ is area-weighted completion. A common floor $\bar{s}_i \geq 0.15Q_i$ protects basic service within the reachable part. If only 20% of a region is reachable, the full-region floor is 3%, not 15%.

$$\begin{aligned} \max_{\mathbf{x}, \mathbf{h}, \mathbf{s}} \quad & 100 \sum_j w_j s_j \\ \text{s.t.} \quad & \left\{ \begin{array}{ll} C_j s_j \leq g_j x_j + u_j h_j, & j \in \mathcal{J}, \\ \sum_j x_j + \sum_{j \in \mathcal{J}_U} \gamma_j h_j \leq H - H_0, & \\ \sum_j h_j \leq U, & \\ \sum_{j \in \mathcal{J}_i} \mu_j s_j \geq 0.15Q_i, & i \in \mathcal{I}, \\ 0 \leq s_j \leq r_j, & j \in \mathcal{J}, \\ x_j, h_j \geq 0, & j \in \mathcal{J}, \\ x_j = 0, & r_j = 0, \\ h_j = 0, & j \notin \mathcal{J}_U. \end{array} \right. \end{aligned} \quad (5.8)$$

The fixed coefficients make this a linear program. Completion is capped at one so excess inspections cannot raise the completion rate above 100%. Human and flight budgets have different units and are linked through operator labor, not added together.

First maximize completion to obtain P^* . Then minimize $H_M = \sum_j x_j + \sum_{j \in \mathcal{J}_U} \gamma_j h_j$ under the same constraints and $100 \sum_j w_j s_j \geq P^*$. All service fractions remain decision variables. This gives the least labor among optimal-rate allocations in the stated model. At the geographic ceiling, positive-weight reachable units must be fully served; zero-weight units still satisfy the regional floor.

5.5 Solution checks and regional allocation

HiGHS solves 1797 variables; independent checks verify capacity, budgets, floors and prohibited allocations. Figure 4 maps monitoring-support labor and drone flight-hours across all 17 reporting regions. The two color scales are normalized independently, so matching colors do not mean equal resource quantities.

Numeric labels identify ENP regions; 3410 is the combined survey unit, P is the main pan and O denotes survey-external areas. Leaders connect displaced labels to their regions. The six stars mark assumed response posts, each reserving 480 personnel-hours outside the regional shading. The maps show pooled monthly work, rather than where employees or aircraft are permanently stationed.

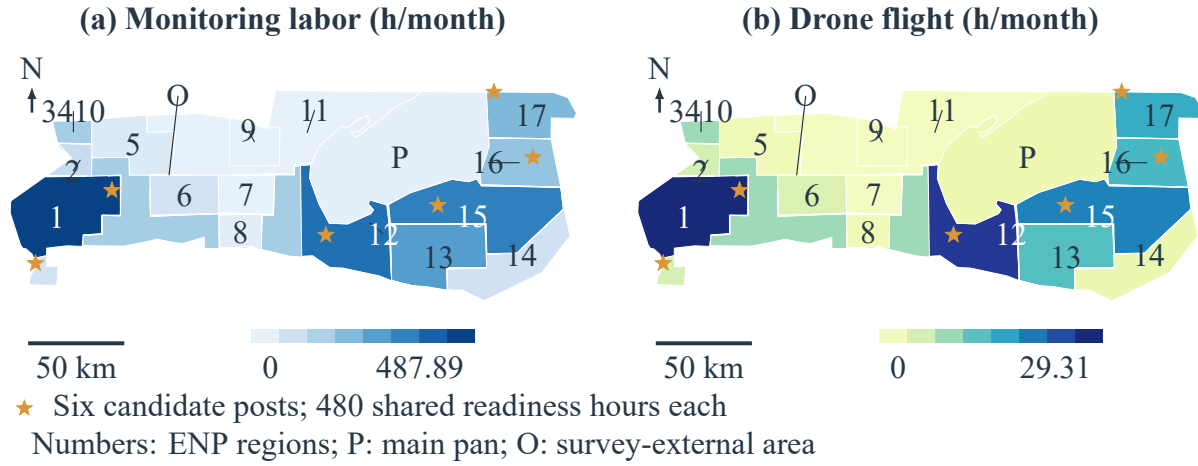


Figure 4: Monthly monitoring-support personnel-hours and drone flight-hours across Etosha's 17 reporting regions.

Table 4 gives the regional totals. Demand uses priority weights; reach and service use area shares within each region. Monitoring labor excludes the shared 2880 readiness hours. ENP1 receives 487.89 support hours and 29.31 flight-hours, while ENP12 receives 385.64 support hours. ENP9 has no qualified response unit; ENP11 has almost no reachable area. The salt pan retains its service floor despite zero measured-component weight.

Table 4: Regional demand, response reach and minimum-labor allocation at the optimal completion rate.

Region	Demand (%)	Reach (%)	Service (%)	Labor (h)	Flight (h)
ENP1	19.06	87.87	87.87	487.89	29.31
ENP12	11.02	92.49	92.49	385.64	26.87
ENP15	9.96	52.33	52.33	352.52	19.90
ENP17	8.84	77.60	77.60	214.11	15.58
ENP3410	8.71	19.21	19.21	155.98	8.38
ENP11	8.05	0.02	0.02	0.26	0.01
ENP16	5.93	84.63	84.63	183.34	14.00
ENP13	5.88	49.87	49.87	282.75	13.08
ENP2	4.50	76.71	76.71	92.86	4.44
ENP5	4.39	5.94	5.94	30.79	0.80
ENP7	3.31	0.19	0.19	0.52	0.02
ENP9	2.90	0.00	0.00	0.00	0.00
ENP14	2.63	11.67	11.67	72.40	2.06
ENP6	2.55	20.57	20.57	60.50	3.62
ENP8	1.54	8.55	8.55	14.93	0.75
Other*	0.72	61.81	61.81	62.52	4.19
Salt pan*	0.00	9.05	1.36	9.11	1.35

* Animal values unknown. Regional labor excludes shared response readiness.

5.6 Strategic comparisons and response posts

Since $w_j \geq 0$ and $s_j \leq r_j$, the geographic upper bound on effective inspection completion is

$$P \leq P_{\text{geo}} = 100 \sum_j w_j r_j. \quad (5.9)$$

The base reaches this bound with 15953.89 personnel-hours and 1055.66 flight-hours unused. Of 599 units, 216 meet the response standard, covering 34.12% of projected area. The 57.26% effective inspection completion rate uses priority weights; the 34.12% figure measures response-eligible area.

We compare the optimized allocation with uniform completion over reachable units and completion proportional to regional demand density W_i/A_i . Costs, resource limits and floors are identical.

Table 5 compares the three rules under identical tasks, weights, response gates and budgets. All reach 57.26% at the baseline and after a 40% labor reduction. Under a separate 3600-hour stress budget, including 2880 readiness hours and 1200 flight-hours, their rates are 16.47%, 19.87% and 29.88%. The advantage appears under scarcity; there is no baseline rate improvement to claim.

Table 5: Completion rates under uniform, demand-based and optimized allocation with identical budgets.

Budget	Labor limit	Uniform	Demand-based	Optimized
Base	21240	57.26	57.26	57.26
Personnel -40%	12744	57.26	57.26	57.26
Tight budget	3600	16.47	19.87	29.88

Removing drones still gives 57.26%, at 6084.98 personnel-hours. Equal crews and checklist times make suitable drone use save off-road walking rather than eliminate personnel.

Table 6 compares three additional response-post candidates. Each adds 480 readiness hours within the original total budgets; routes, response eligibility and inspection costs are rebuilt, while weights stay fixed. This isolates the effect of improved response access without increasing the labor allowance.

Table 6: Additional response-post candidates tested within the original resource budgets.

Layout	Readiness (h)	Rate (%)	Total labor (h)	Flight (h)
Original six posts	2880	57.26	5286.11	144.34
Add ENP11	3360	58.86	5888.64	154.05
Add ENP3410	3360	58.95	5825.11	151.03
Add ENP5	3360	64.27	6202.08	181.77

ENP5 raises completion from 57.26% to 64.27%, using 6202.08 personnel-hours. Its 7.01-point gain is the largest among the three candidates, rather than a global siting optimum. Facilities, permissions and access need verification before adoption.

6. Seasonal Service and Workforce Requirements

Question 3 asks how service changes through time and how many people are needed to maintain a chosen level. We extend the basic inspections with seasonal work, then convert minimum workload into personnel requirements.

6.1 Target, months and additional tasks

We use the same weights, inspection standard, routes, response locations and technical eligibility as Question 2. The target is its achieved optimum of 57.26%, using the unrounded percentage value $\eta = 57.26016128$; its ecological sufficiency is not established by being mathematically optimal.

Each planning month has 30 days and 120 effective personnel-hours per person. Reference availability is 177 people and 1200 monthly flight-hours. Assignments may change between months. Existing equipment, operators and response staff cannot be charged twice for concurrent tasks.

The dry-season classification is May–October in the cited study; the official fire strategy distinguishes May–July and August–November operating windows [9, 6]. These sources support the timing, not the numerical task frequencies. Table 7 gives the assumed frequencies. Wet months use two maintenance visits per point and dry months use four; extra fire screening rises from zero to two and then four checks per 100 km² of burnable habitat. November combines wet maintenance with the late-fire rule. These additional tasks use the shared resources without redefining the base inspection score.

Table 7: Seasonal visit frequencies and extra fire checks under the medium service rules.

Months	Water visits/point	Extra fire checks*
Jan–Apr, Dec	2	0
May–Jul	4	2
Aug–Oct	4	4
Nov	2	4

*Checks per 100 km² of burnable habitat per month; on-site maintenance is 0.5 hour/visit in normal months.

Extra fire checks are distinct tasks added to the base checklist. Let A_j^{fuel} be the unit's support area multiplied by its region's observed burnable-habitat fraction. Then

$$D_{jt} = k_t \frac{A_j^{\text{fuel}}}{100} r_j. \quad (6.1)$$

The same inspection protocol and cost coefficients apply. These are screening tasks, not firefighting or prescribed burning. The factor r_j restricts credited tasks to the existing response range; excluded demand is separately recorded.

At the late-fire rate, 275.76 additional checks are supported inside that range, while 428.45 fall outside it. Adding personnel to the same six-post layout cannot guarantee those excluded tasks. We retain this gap rather than claiming park-wide fire coverage.

6.2 Borehole maintenance as ground-only work

A published 2013 field table contains 42 water-point records; 17 are classified as boreholes [7]. We use those 17 mapped points as a historical maintenance sample. It is not a complete current operational inventory. Natural springs are not assigned pumping-equipment work. Ministry records support solar-powered borehole equipment and maintenance as relevant activities [8].

Each visit includes a directed road round trip from a candidate post, any required walking, preparation and on-site inspection. We assume two people travel together. Let T_ℓ^{rt} be elapsed round-trip travel time, f_t visits per point and τ_t on-site time. Monthly personnel work is

$$\begin{aligned} a_{\ell t}^W &= 2(T_\ell^{\text{rt}} + \tau_t + 1/12), \\ W_t &= f_t \sum_{\ell \in \mathcal{L}} a_{\ell t}^W. \end{aligned} \quad (6.2)$$

The 1/12 preparation term is five minutes. Physical maintenance remains ground work. Table 8 compares the three rules.

Summed round-trip travel is 42.87565 elapsed hours. The regular dry-month workload is

$$W_{\text{dry}} = 4(2) [42.87565 + 17(0.5 + 1/12)] = 422.34 \text{ personnel-hours}. \quad (6.3)$$

Major repairs, unlisted points and route-sharing need updated records. The longest visit takes about 6.39 elapsed hours; a single task fitting an eight-hour day does not prove a roster.

Table 8: Monthly maintenance workloads for the 17 historical boreholes under three service rules.

Rule	Visits/point	On-site hours	Monthly labor
Wet regular	2	0.5	211.17
Dry regular	4	0.5	422.34
Drought pressure	8	1.0	980.68

Two regular visits per borehole require 211.17 personnel-hours monthly. The drought-maintenance rule assumes eight visits and one hour on site, giving 980.68 personnel-hours. Each visit is a separate round trip, so the calculation claims no combined-route savings; the increases reflect assumed service rules, rather than recorded faults or measured shortages.

6.3 Shared monthly constraints

For each unit and month, (x_{jt}, h_{jt}, s_{jt}) serve the base inspection protocol. Extra fire work uses ground hours y_{jt} and flight-hours v_{jt} . These are separate task quantities with shared personnel and flight budgets. Where a technology is unavailable, its capacity term and investment are zero.

$$\begin{aligned} C_j s_{jt} &\leq g_j x_{jt} + u_j h_{jt}, \\ D_{jt} &\leq g_j y_{jt} + u_j v_{jt}, \\ \sum_j (h_{jt} + v_{jt}) &\leq U, \\ H_t &= H_0 + W_t + \sum_j (x_{jt} + y_{jt}) + \sum_{j \in \mathcal{J}_U} \gamma_j (h_{jt} + v_{jt}). \end{aligned} \tag{6.4}$$

The remaining conditions retain $0 \leq s_{jt} \leq r_j$, nonnegative investments, technology eligibility and the regional floor

$$\sum_{j \in \mathcal{J}_i} \mu_j s_{jt} \geq 0.15 \sum_{j \in \mathcal{J}_i} \mu_j r_j. \tag{6.5}$$

Only basic checks enter $P_t = 100 \sum_j w_j s_{jt}$. Required seasonal tasks use the same budgets without changing that score. Readiness is charged once; monthly feasibility still requires date, qualification and concurrent-alert checks.

6.4 Forward service and inverse workforce models

Two uses of the same feasible task set answer different questions. With N available people, the forward model finds the best monthly service. With a fixed service target, the inverse model finds the least total personnel work:

$$\begin{aligned} P_t^*(N, U) &= \max_{\mathbf{z}_t \in \mathcal{F}_t(U), H_t \leq 120N} 100 \sum_j w_j s_{jt}, \\ H_t^{\min}(\eta, U) &= \min_{\mathbf{z}_t \in \mathcal{F}_t(U), P_t \geq \eta} H_t. \end{aligned} \tag{6.6}$$

Here $\mathbf{z}_t$ collects all five variable blocks and $\mathcal{F}_t(U)$ contains the shared constraints. The inverse model removes the reference labor ceiling so that a shortage can be expressed as a required workforce instead of only returning infeasibility. As in revised Question 2, it allows service fractions to vary among equal-rate allocations.

$$\begin{aligned} N_t &= \left\lceil \frac{H_t^{\min}}{120} \right\rceil, \\ N_{\text{year}} &= \max_{t=1, \dots, 12} N_t. \end{aligned} \tag{6.7}$$

With seasonal tasks removed, the independently constructed inverse model returns 5286.11 personnel-hours, matching revised Question 2. This cross-check confirms that both chapters use the same optimal-rate, minimum-labor objective.

Figure 5 separates response readiness, basic inspection support, extra fire checks and borehole maintenance. The reference capacity of 177 people is outside the plotted workload scale. The maximum monthly requirement, rather than the sum of monthly headcounts, estimates a constant workforce for the busiest month. It does not certify qualifications, exact simultaneous coverage or the completeness of the task inventory.

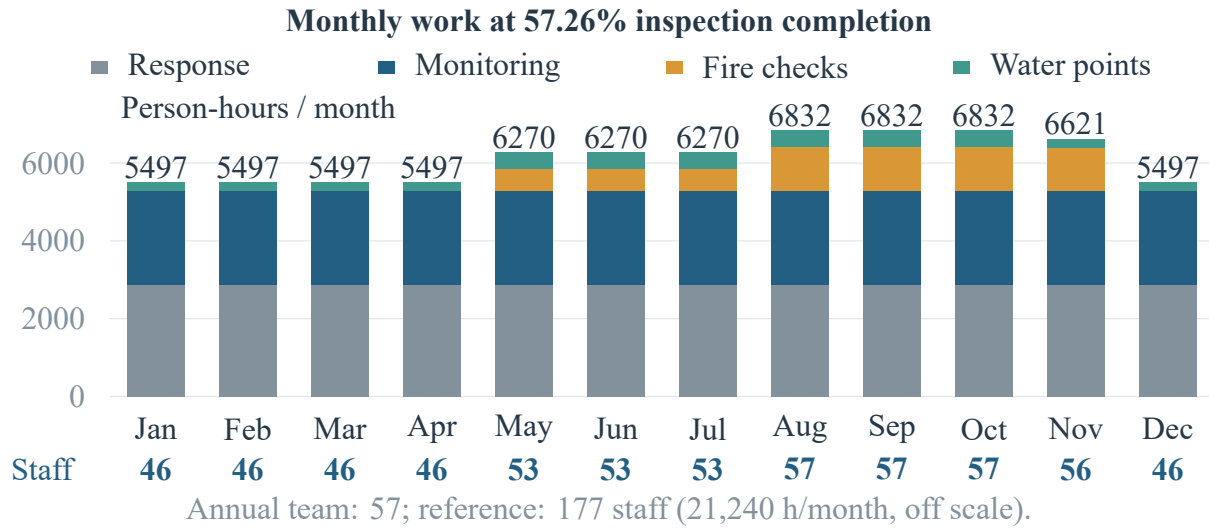


Figure 5: Monthly workload by task for maintaining the baseline inspection target.

6.5 Normal-season results and feasibility

Table 9 reports that all months maintain the target with the 177-person reference and 1200 flight-hours. The least-work solutions need fewer personnel equivalents because only specified tasks and the existing response range are modeled.

The peak occurs in August–October. Its work separates into four parts:

$$H_{\text{peak}}^{\min} = 2880 + 2406.11 + 1123.38 + 422.34 = 6831.83 \text{ personnel-hours.} \quad (6.8)$$

These are readiness, base inspection labor, extra fire inspection labor and borehole maintenance. The flight requirement is 210.61 hours, well below 1200. Rounding $H_{\text{peak}}^{\min}/120$ gives 57 people. A check with 56 people returns 56.73, below the target, under the same monthly constraints.

Table 9: Minimum monthly workload and personnel equivalents under the medium seasonal rules.

Months	Personnel-hours	People	Flight-hours	Rate at 177 staff (%)
Jan–Apr, Dec	5497.28	46	144.34	57.26
May–Jul	6270.14	53	177.48	57.26
Aug–Oct	6831.83	57	210.61	57.26
Nov	6620.66	56	210.61	57.26

This is a task-specific monthly minimum, not an operational staffing recommendation. Specialist, night and simultaneous incident work are omitted.

Independent task-capacity, qualification, budget and floor checks verify the saved plans. Field costs, permissions and checklist effectiveness require separate calibration.

Extra seasonal work uses spare capacity without expanding the fixed response range. Raising the basic target requires improved access or deployment.

6.6 Drought work and service-standard sensitivity

The drought-maintenance case doubles dry-month visits and on-site time. Its peak is 7390.17 personnel-hours, or 62 equivalents, with 210.61 flight-hours. Table 10 compares alternative rules.

Table 10: Peak personnel requirements under alternative seasonal and drought-maintenance rules.

Rule	Normal people	Normal hours	Drought people	Drought hours
Low	51	6041.97	53	6270.14
Medium	57	6831.83	62	7390.17
High	71	8513.55	77	9207.89

The low, medium and high rules require normal peak equivalents of 51, 57 and 71; their drought-maintenance counterparts require 53, 62 and 77. These are consequences of prescribed visit and screening frequencies, rather than forecasts or confidence limits. Fewer personnel under a relaxed rule do not establish equal ecological protection.

7. Sensitivity and Scenario Analysis

Figure 6 tests joint budgets. Reducing baseline personnel-hours by 40% still gives 57.26% effective inspection completion. Lower resource budgets are tested with weights and tasks fixed. All 127 feasible saved solutions pass independent checks; four infeasible cases are excluded rather than plotted as zero completion. Matrix rows specify personnel at 120 effective hours each, and columns specify flight-hours. Each cell is reoptimized for the same medium peak-month tasks, including mandatory maintenance and extra fire screening.

Mid-level seasonal peak: joint personnel and flight capacity						
Staff	0	80	160	200	240	1200
45	20.3%	33.8%	37.2%	37.2%	37.2%	37.2%
55	44.7%	52.0%	55.2%	55.7%	55.7%	55.7%
57	47.7%	54.1%	57.0%	57.3%	57.3%	57.3%
62	53.6%	57.3%	57.3%	57.3%	57.3%	57.3%
67	57.3%	57.3%	57.3%	57.3%	57.3%	57.3%
177	57.3%	57.3%	57.3%	57.3%	57.3%	57.3%

Columns: flight-hours/month; cells: effective inspection completion (%).

Target 57.26%; 177 staff and 1200 flight-hours are the reference.

Figure 6: Peak-month completion rates under joint personnel and drone-flight budgets.

At the medium peak, 57 people with 200 flight-hours maintain the target; 57 with 160 hours reach 56.97%. With no drones, the target needs 67 people. With 59 people and 120 flight-hours, reallocation reaches 57.24%, compared with 57.14% under the retained spatial service rule. Under the specified drought-maintenance pressure at the same low budgets, the gain from reallocation is 6.83 percentage points, but the target is still missed.

For the same 68 borehole visits and 422.34 monthly personnel-hours, staggered dates reduce the maximum gap from 27 to 8 days and the daily water-work peak from 105.58 to 15.48 hours. This water-only schedule is not a full roster or proof of a strict 7.5-day rule.

At road speeds of 20 and 40 km/h, the completion-rate ceilings are 35.76% and 64.69%. Extending response to three hours gives 80.64% under a changed service standard. These are conditional scenarios, not measured speed ranges or event probabilities.

7.1 Screening protocol and grid sensitivity

Let e be drone screening capacity relative to the baseline checklist and p_g the required ground-check share. Replace task capacity by $C_j s_j \leq x_j/a_j + e h_j/b_j$ and, where required, impose $x_j/a_j \geq p_g C_j s_j$. The baseline has $e = 1$ and $p_g = 0$. Table 11 tests these assumed protocols, rather than measured detection probabilities.

Table 11: Technology allocation under alternative drone-efficiency and minimum-ground-share assumptions.

Protocol	Rate (%)	Total labor (h)	Ground (h)	Flight (h)
Equivalent checklist	57.26	5286.11	0.00	144.34
Drone efficiency 75%	57.26	5786.37	1731.55	104.27
Drone efficiency 50%	57.26	6011.98	2964.54	33.35
Ground minimum 10%	57.26	5366.13	320.63	129.91
Ground minimum 25%	57.26	5486.15	801.56	108.26

At 75% drone efficiency, ground screening returns at 1731.55 personnel-hours and total work rises to 5786.37. A 25% ground-check requirement uses 801.56 ground hours and 5486.15 total hours. All tested protocols still reach 57.26% with ample budgets. Thus zero baseline ground screening depends on checklist equivalence; operators and responders remain necessary.

Table 12 changes grid size while holding area-based task density, budgets and response rules fixed. The 5, 10 and 15 km grids contain 1554, 599 and 392 connected units. Refinement changes representative locations without creating extra aggregate check demand.

Table 12: Sensitivity of completion rates and workload to inspection-grid size.

Grid side (km)	Units	Rate (%)	Total labor (h)
5	1554	56.39	5279.44
10	599	57.26	5286.11
15	392	59.99	5340.08

The three grids yield 56.39%, 57.26% and 59.99% completion. Relative to the baseline, the finer result is 0.87 points lower and the coarser result is 2.73 points higher. Differences arise from representative locations and response qualification near the deadline, rather than statistical sampling uncertainty.

7.2 Weight sensitivity

Reoptimization keeps other conditions fixed. Entropy measures spatial differences; CRITIC uses variation and indicator overlap. Table 13 separates each plan’s own-weight score from its score under the original entropy weights. At the ample baseline, all four allocations are identical and original-weight completion stays at 57.26%, despite changes in the scoring scale. This common evaluation prevents a changed definition of importance from being read as improved service.

For any saved allocation $\mathbf{s}$, the common rate is

$$P_{\text{ref}}(\mathbf{s}) = 100 \sum_j w_j^{\text{base}} s_j.$$

The same base weights evaluate every plan, while the own-weight rate uses that plan’s chosen weights. The reference calculation therefore measures deployment differences without changing the evaluation rule.

The separate scarcity trial allows 3600 personnel-hours, including 2880 readiness hours, and 1200 flight-hours. Up to 9 units change completion relative to its entropy solution, while rates evaluated with the original weights span only 29.86–29.88%. Deployment responds to priorities, but the completion-rate effect is small in these tests.

Perturbing specially protected category preferences by $\pm 20\%$, with reciprocal comparisons and recalculated weights, preserves matrix consistency and base deployment. Tight-budget completion-rate losses stay below 0.002 percentage points. All cases pass independent capacity, eligibility, regional-floor and budget checks; robustness applies to these tested choices.

Table 13: Allocation scores under alternative responsibility weights and the common reference weights.

Weight choice	Animal (%)	Own weights	Original weights	Tight budget
Entropy baseline	64.56	57.26	57.26	29.88
Equal components	50.00	53.18	57.26	29.87
Animal emphasis	90.00	64.40	57.26	29.88
CRITIC alternative	44.80	51.72	57.26	29.86

8. Strengths, Limitations and Practical Use

We assess computational strengths and the evidence needed for operational use.

8.1 Strengths

The task-based framework separates budget scarcity from response access. Shared personnel and distinct flight constraints make resource use explicit; inverse workloads explain seasonal demand, with independent checks enabling reproducibility.

8.2 Weaknesses and Limitations

Some species and facility records are historical and incomplete, so the responsibilities we can measure are limited. Checklist and cost assumptions affect how service completion is interpreted. Mapped roads do not guarantee legal or seasonal access, and total monthly working hours cannot ensure that qualified crews are available at every time. The 17-point maintenance sample may omit current facilities or repair tasks.

For these reasons, the calculated 57.26% optimum and the 57 or 62 personnel equivalents do not establish sufficient ecological protection or exact park staffing needs. Table 14 links each limitation to its consequences and the evidence still needed. Independent mathematical checks confirm feasibility and numerical consistency within the specified model; incident, survey, habitat and field-work records are needed to evaluate real outcomes.

Table 14: Operational limitations and the evidence required for field validation.

Limit	Consequence and data needed
Historical / missing objects	Current priority-species locations and surveys are needed; six measured groups do not certify all wildlife.
Service proxies and assumptions	Calibrate check frequency, category priorities, fire exposure and ground/drone checklist equivalence.
Maps and permissions	Verify roads, barriers, seasons and emergency access; a mapped route is not confirmed permission.
Monthly pooled work	Check dates, qualifications, integer visits and concurrent incidents before calling a roster feasible.
Incomplete maintenance inventory	Update borehole status, tasks and repair duration; the 17-point sample cannot define all current work.
Service versus ecological outcomes	Validate loss, habitat and population outcomes independently; optimal service does not prove adequate protection.

Before deployment, verify access and task effectiveness, update records, retain safeguards for missing objects, and validate ecological outcomes independently.

9. Adaptation to Parks on Other Continents

We examine Chitwan National Park in Asia and Yellowstone National Park in North America, rebuilding local geography and testing how priorities affect deployment.

9.1 Shared structure and local data

Priority setting, task capacity, response gates, readiness budgets and workforce conversion can remain. Species, habitats, seasonal routes, task protocols, permissions and operating costs must be rebuilt locally. Etosha’s animal weights and thirty-drone assumption are not transferred.

Figure 7 shows the published geography and candidate response locations. Chitwan uses a public protected-area polygon, mapped roads and geographic anchors; Yellowstone uses NPS geography, public roads and station locations. Stars denote candidates for the trial, rather than observed staffing assignments. Metric projections support travel calculations, while local speeds, task times and permissions remain assumptions.

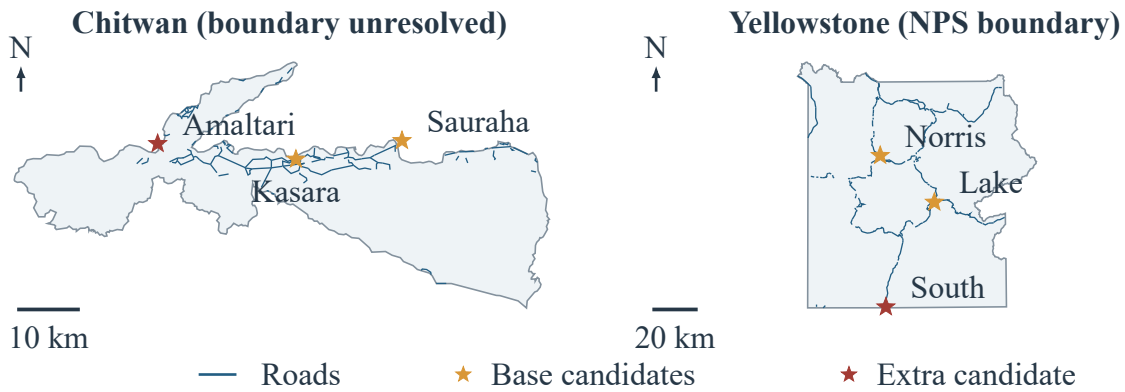


Figure 7: Published geography and candidate response posts for Chitwan and Yellowstone.

Table 15 summarizes the boundaries, roads and WorldCover 2021 classification. The trials use land-area weights $w_j = A_j^{\text{land}} / \sum_k A_k^{\text{land}}$ and $C_j = 8A_j^{\text{land}}/100$. Drone investment is zero because authorization and task equivalence are not calibrated. The 15% floor applies in each reachable cell where reporting regions are unavailable; Etosha keeps regional floors. This changes the safeguard scale locally, and tests land service rather than complete species protection.

Table 15: Spatial inputs used in the Chitwan and Yellowstone land-inspection trials.

Input	Chitwan polygon	Yellowstone NPS
Land service area (km ²)	1175.65	8470.07
Forest share (%)	91.11	62.93
Mapped park roads (km)	248.29	619.60
Units / grid side	1376 / 1 km	1520 / 2.5 km

Boundary restriction. Chitwan’s 1214.06 km² polygon exceeds the official 952.63 by 27.44%. Its 1213.00 km² geodesic area matches source GIS; projection accounts for only 0.087%. The official 2020 GIS report also gives 952.63 [16]. The boundary/reporting convention remains unresolved. Yellowstone uses NPS geography [13].

9.2 Conditional trials and expected deployment changes

For each park, fix its target at 90% of its own two-post, 30 km/h response-limited ceiling:

$$\begin{aligned}
 P_{\text{geo}}^{(p)} &= 100 \sum_j w_j^{(p)} r_j^{(p)}, \\
 \eta^{(p)} &= 0.90 P_{\text{geo,base}}^{(p)}.
 \end{aligned}
 \tag{9.1}$$

Table 16 retains fixed targets. Two posts reserve 960 personnel-hours; a third adds 480. Local specialist tasks remain unmeasured.

Table 16: Conditional land-inspection workloads under local speed and response-post scenarios.

Case	Setting	Ceiling	Target	Labor (h)	People
C0	Two posts, 30 km/h	37.57	33.81	1085.15	10
C1	20 km/h	30.66	33.81	Infeasible	–
C2	Third post added	45.58	33.81	1550.39	13
C3	40 km/h	42.12	33.81	1069.56	9
Y0	Two posts, 30 km/h	18.72	16.85	1541.63	13
Y1	20 km/h	12.02	16.85	Infeasible	–
Y2	Third post added	21.60	16.85	1973.23	17
Y3	40 km/h	24.48	16.85	1467.97	13

At 20 km/h, both parks' fixed targets exceed their new response ceilings, so adding people alone cannot meet the target. A third post expands response reach but adds readiness work. Higher speed is a sensitivity assumption, rather than a driving recommendation. These land-inspection trials do not determine actual park staffing.

Calibrate Chitwan's forest, river and community tasks, and Yellowstone's visitor and winter tasks; closures do not establish emergency permissions [14].

Half-grid tests change caps by -0.13 and -0.20 points; finer land-cover sampling changes forest shares by -0.002 and -0.056 points. Ten numerical cases pass; Chitwan's boundary check fails. Staffing equivalents remain conditional.

9.3 A local historical-priority trial

The NPS 2022 wolf-pack activity files contain ten published records and nine distinct geometries. Their clipped union covers 4093.87 km^2 ; it describes historical activity extent, rather than current density, measured threat or official management priorities [17].

Table 17 mixes normalized historical overlap, adjusted by existing cell land fraction, with land-area responsibility: $w_j^{(\lambda)} = (1 - \lambda)w_j^{\text{land}} + \lambda w_j^{\text{hist}}$. Each trial preserves Y0's score under its new weights and minimizes labor with roads, tasks and service floors unchanged. At a 50% historical share, 61 units change, labor is 1474.68 hours, and land-weight service falls from 16.85% to 15.12%. This is a priority trade-off, rather than an ecological gain. The adjacent appendix gives the complete comparison.

Appendix: supplementary results

Table 17 supplies the complete historical-priority comparison discussed in Section 9.3. All road, task and response inputs are held fixed; the priorities and retained targets are declared scenario choices.

Table 17: Yellowstone historical-priority scenarios with roads and inspection tasks held fixed.

Wolf share (%)	New-rule rate	Land-rule rate	Labor (h)	Changed units
0	16.85	16.85	1541.63	0
50	20.04	15.12	1474.68	61
75	21.64	13.04	1413.44	102

The following reproducibility entry calls the same allocation solver using saved real inputs. It reports the rate, minimum personnel-hours and flight-hours, without supplying synthetic observations. Its exact source is also included in the support archive.

```
from pathlib import Path
import json
import sys
```

```
root = Path.cwd()
sys.path.insert(0, str(root / 'scripts/modeling'))
from build_question2 import solve
```

```
path = root / 'output/question2/q2_model_inputs.json'
data = json.loads(path.read_text(encoding='utf-8'))
result = solve(data)
assert result['success']
for name in ['score', 'total_person_hours', 'drone_flight_hours']:
    print(name, round(result[name], 2))
```

Letter to IMMC

To the International Mission for Monitoring Conservation

Subject: A practical protection plan for Etosha and other large reserves

Dear IMMC Decision-Makers,

We recommend connecting monitoring to timely ground response. Our Etosha study shows that staff and drones can remain unused while important locations are too distant for effective follow-up.

Our approach divides the park into planning areas, uses historical animal surveys and burnable habitat to guide priorities, and calculates task and travel time. Routes and response posts determine which checks receive support. Drones can save walking time in suitable open locations, but people must still transport and operate them and follow up reported problems.

The planning baseline uses 60% of 295 reference employees, or 177 available people at 120 effective hours each, thirty drones and two-hour ground arrival. These assumptions do not describe verified current staffing or equipment.

The highest effective inspection completion rate under these conditions is 57.26%. The minimum-work month uses about 5286 personnel-hours, including readiness, and 144 flight-hours. Both are well below the resource limits. This rate combines task completion and ground-response eligibility using conservation-priority weights. Wildlife losses, population health and habitat conditions need separate evaluation.

Among three additional-post candidates tested within the same budgets, ENP5 raises the completion rate to 64.27%. It is a direction for field assessment, not a confirmed global optimum. Check routes, permissions, site conditions and response needs before adoption.

Seasonal work also needs preparation. Additional fire checks and borehole maintenance raise the modeled normal peak requirement to 57 people. The specified drought-maintenance scenario raises it to 62. Both fit within the 177-person reference, but cover only the selected daytime tasks, response range and historical maintenance sample.

Begin with a limited field trial: record travel, inspection, drone use and follow-up time; compare checklist completion by ground and drone-assisted teams; update the maintenance inventory; and verify access. Use these records to revise routes, response locations and task assumptions.

Turn the monthly plan into a calendar. Stagger borehole visits to avoid long service gaps, check daily capacity and retain qualified response staff. Thirty drones do not imply thirty simultaneous flights; operator, readiness and maintenance work all need time.

Trials in Chitwan and Yellowstone show how the structure can be adapted using local spatial records and operating assumptions. Chitwan's boundary disagrees with the official area, so its trial cannot support a calibrated staffing recommendation. Yellowstone requires information on visitor interactions and winter access. Neither trial establishes actual park staffing.

At Etosha, records come from different years, and some priority species, including black rhinoceros, lack detailed distributions. The model assumes occasional alerts rather than several serious simultaneous incidents. Specialist and night work may add personnel needs; a high inspection completion rate cannot resolve these gaps.

We propose three connected actions: validate field assumptions; examine response locations that could close important access gaps; and revise monthly assignments using verified records. Retain basic service in less prominent areas and dedicated safeguards for priority species. Measure progress through completed checks, reliable follow-up and independently observed wildlife and habitat outcomes.

Personnel, equipment and access should be planned together. The framework provides a transparent starting point, with decisions revised as reliable evidence improves.

Yours faithfully,

The Modeling Team

Team 510105201001220151

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Report on Use of AI

Tool and identification. OpenAI Codex, based on the GPT-6 model family, was used in this work session [15]. The exact deployed model version was not exposed in the session metadata. Python/SciPy and XeLaTeX performed numerical optimization and document compilation; those outputs were checked separately from generated prose.

Uses. The assistant helped explain existing formulas, distinguish sourced records from planning assumptions, revise resource parameters specified by the team, synchronize seasonal calculations, shorten chapter drafts, translate the integrated manuscript into English, draft the nontechnical letter and produce numbered figures and tables. Parallel assistants worked on bounded chapter, data-synchronization and visualization tasks.

Representative instructions. The team requested: (1) explain each stage and formula in the existing Question 2 document; (2) keep other parameters unchanged, make 60% of personnel available and use thirty drones, then shorten Question 2 to eight pages with simple terminology; (3) retrieve remote updates, revise Question 3 to six pages, revise Question 6 similarly, and integrate Questions 1–6, the template sections and a letter into an English paper of at most 24 pages. The team supplied the control number used here and subsequently requested a multi-species scope, correction of the two-stage objective and feasibility-based manuscript revision.

Verification. Reported Question 2–4 allocations were recomputed or checked against saved inputs and explicit constraints. The cross-park chapter uses the retrieved real-space results and retains its failed Chitwan boundary check. Scripts verify units, capacities, budgets, prohibitions, floors, totals, references, figure/table labels and final page count. The present revision includes renewed numerical checks and PDF layout review; the corresponding audit records are retained with the source. A computational check is not a validation of real ecological outcomes or a detailed roster.

Limits and responsibility. Generated wording and model choices require team review. The manuscript identifies historical data, scenario assumptions and uncalibrated inputs; it makes no claim that AI supplied current animal distributions, verified staffing inventories, ecological-loss effects or authorized access permissions. Sources are cited separately from the team's calculations. The team remains responsible for accuracy, originality and compliance with the competition's disclosure rules.

The uploaded template's obsolete year, example notation, sample identifiers and unfinished claims were replaced with the current problem, confirmed team number and verified results. The project retains source data, model inputs, allocation arrays, code, provenance and numerical/layout audit records for review.